

"Thanks, I guess." He twists the throttle up once and Scott feels, but does not hear, the power of the engine. This baby is so efficient it doesn't waste power by making noise. "Say hi to your brand-new niece," the guy says, and then lets go the clutch. The spokes flex and gather themselves and the bike springs forward out of the lot, seeming to jump off its electric paws. He cuts right across the parking lot of the neighboring NeoAquarian Temple Franchise and pulls out onto the road. About half a second later, the guy with the swords is a dot on the horizon. Then he's gone. Northbound.

Chapter 36

Until a man is twenty-five, he still thinks, every so often, that under the right circumstances he could be the baddest motherfucker in the world. If I moved to a martial-arts monastery in China and studied real hard for ten years. If my family was wiped out by Colombian drug dealers and I swore myself to revenge. If I got a fatal disease, had one year to live, devoted it to wiping out street crime. If I just dropped out and devoted my life to being bad.

Hiro used to feel that way, too, but then he ran into Raven. In a way, this is liberating. He no longer has to worry about trying to be the baddest motherfucker in the world. The position is taken. The crowning touch, the one thing that really puts true world-class badmotherfuckerdom totally out of reach, of course, is the hydrogen bomb. If it wasn't for the hydrogen bomb, a man could still aspire. Maybe find Raven's Achilles' heel. Sneak up, get a drop, slip a mickey, pull a fast one. But Raven's nuclear umbrella kind of puts the world title out of reach.

Which is okay. Sometimes it's all right just to be a little bad. To know your limitations. Make do with what you've got.

Once he maneuvers his way onto the freeway, aimed up into the mountains, he goggles into his office. Earth is still there, zoomed in tight on the Raft. Hiro contemplates it, superimposed in ghostly hues on his view of the highway, as he rides toward Oregon at a hundred and forty miles per hour.

From a distance, it looks bigger than it really is. Getting closer, he can see that this illusion is caused by an enveloping, self-made slick/cloud of sewage and air pollution, fading out into the ocean and the atmosphere.

It orbits the Pacific clockwise. When they fire up the boilers on the Enterprise, it can control its direction a little bit, but real navigation is a practical impossibility with all the other shit lashed onto it. It mostly has to go where the wind and the Coriolis effect take it. A couple of years ago, it was going by the Philippines, Vietnam, China, Siberia, picking up Refus. Then it swung up the Aleutian chain, down the Alaska panhandle, and now it's gliding past the small town of Port Sherman, Oregon, near the California border.

As the Raft moves through the Pacific, riding mostly on ocean currents, it occasionally sheds great hunks of itself. Eventually, these fragments wash up in some place like Santa Barbara, still lashed together, carrying a payload of skeletons and gnawed bones.

When it gets to California, it will enter a new phase of its life cycle. It will shed much of its sprawling improvised bulk as a few hundred thousand Refus cut themselves loose and paddle to shore. The only Refus who make it that far are, by definition, the ones who were agile enough to make it out to the Raft in the first place, resourceful enough to survive the agonizingly slow passage through the arctic waters, and tough enough not to get killed by any of the other Refus. Nice guys, all of them. Just the kind of people you'd like to have showing up on your private beach in groups of a few thousand.

Stripped down to a few major ships, a little more maneuverable, the Enterprise then will swing across the South Pacific, heading for Indonesia, where it will turn north again and start the next cycle of migration.

Army ants cross mighty rivers by climbing on top of each other and clustering together into a little ball that floats. Many of them fall off and sink, and naturally the ants on the bottom of the ball drown. The ones who are quick and vigorous enough to keep clawing their way to the top survive. A lot of them make it across, and that's why you can't stop army ants by dynamiting the bridges. That's how Refus come across the Pacific, even though they are too poor to book passage on a real ship or buy a seaworthy boat. A new wave washes up onto the West Coast every five years or so, when the ocean currents bring the Enterprise back.

For the last couple of months, owners of beachfront property in California have been hiring security people, putting up spotlights and antipersonnel fences along the tide line, mounting machine guns on their yachts. They have all subscribed to CIC's twenty-four-hour Raft Report, getting the latest news flash, straight from the satellite, on when the Latest contingent of twenty-five thousand starving Eurasians has cut itself loose from the Enterprise and started dipping its myriad oars into the Pacific, like ant legs.

"Time to do more digging," he tells the Librarian. "But this is going to have to be totally verbal, because I'm headed up I-5 at some incredible speed right now, and I have to watch out for slow-moving bagos and stuff."

"I'll keep that in mind," the voice of the Librarian says into his earphones. "Look out for the jackknifed truck south of Santa Clarita. And there is a large chuckhole in the left lane near the Tulare exit."

"Thanks. Who were these gods anyway? Did Lagos have an opinion on that?"

"Lagos believed that they might have been magicians -- that is, normal human beings with special powers -- or they might have been aliens."

"Whoa, whoa, hold on. Let's take these one at a time. What did Lagos mean when he talked about 'normal human beings with special powers'?"

"Assume that the nam-shub of Enki really functioned as a virus. Assume that someone named Enki invented it. Then Enki must have had some kind of linguistic power that goes beyond our concept of normal."

"And how would this power work? What's the mechanism?"

"I can only give you forward references drawn by Lagos."

"Okay. Give me some."

"The belief in the magical power of language is not unusual, both in mystical and academic literature. The Kabbalists -- Jewish mystics of Spain and Palestine -- believed that super-normal insight and power could be derived from properly combining the letters of the Divine Name. For example, Abu Aharon, an early Kabbalist who emigrated from Baghdad to Italy, was said to perform miracles through the power of the Sacred Names."

"What kind of power are we talking about here?"

"Most Kabbalists were theorists who were interested only in pure meditation. But there were so-called 'practical Kabbalists' who tried to apply the power of the Kabbalah in everyday life."

"In other words, sorcerers."

"Yes. These practical kabbalists used a so-called 'archangelic alphabet,' derived from first-century Greek and Aramaic theurgic alphabets, which resembled cuneiform. The Kabbalists referred to this alphabet as 'eye writing,' because the letters were composed of lines and small circles, which resembled eyes."

"Ones and zeroes."

"Some Kabbalists divided up the letters of the alphabet according to where they were produced inside the mouth."

"Okay. So as we would think of it, they were drawing a connection between the printed letter on the page and the neural connections that had to be invoked in order to pronounce it."

"Yes. By analyzing the spelling of various words, they were able to draw what they thought were profound conclusions about their true, inner meaning and significance."

"Okay. If you say so."

"In the academic realm, the literature is naturally not as fanciful. But a great deal of effort has been devoted to explaining Babel. Not the Babel event -- which most people consider to be a myth -- but the fact that languages tend to diverge. A number of linguistic theories have been developed in an effort to tie all languages together."

"Theories Lagos tried to apply to his virus hypothesis."

"Yes. There are two schools: relativists and universalists. As George Steiner summarizes it, relativists tend to believe that language is not the vehicle of thought but its determining medium. It is the framework of cognition. Our perceptions of everything are organized by the flux of sensations passing over that framework. Hence, the study of the evolution of language is the study of the evolution of the human mind itself."

"Okay, I can see the significance of that. What about the universalists?"

"In contrast with the relativists, who believe that languages need not have anything in common with each other, the universalists believe that if you can analyze languages enough, you can find that all of them have certain traits in common. So they analyze languages, looking for such traits."

"Have they found any?"

"No. There seems to be an exception to every rule."

"Which blows universalism out of the water."

"Not necessarily. They explain this problem by saying that the shared traits are too deeply buried to be analyzable."

"Which is a cop out."

"Their point is that at some level, language has to happen inside the human brain. Since all human brains are more or less the same --"

"The hardware's the same. Not the software."

"You are using some kind of metaphor that I cannot understand."

Hiro whips past a big Airstream that is rocking from side to side in a dangerous wind coming down the valley.

"Well, a French-speaker's brain starts out the same as an English-speaker's brain. As they grow up, they get programmed with different software -- they learn different languages."

"Yes. Therefore, according to the universalists, French and English -- or any other languages -- must share certain traits that have their roots in the 'deep structures' of the human brain. According to Chomskyan theory, the deep structures are innate components of the brain that enable it to carry out certain formal kinds of operations on strings of symbols. Or, as Steiner paraphrases Emmon Bach: These deep structures eventually lead to the actual patterning of the cortex with its immensely ramified yet, at the same time, 'programmed' network of electrochemical and neurophysiological channels."

"But these deep structures are so deep we can't even see them?"

"The universalists place the active nodes of linguistic life -- the deep structures -- so deep as to defy observation and description. Or to use Steiner's analogy: Try to draw up the creature from the depths of the sea, and it will disintegrate or change form grotesquely."

"There's that serpent again. So which theory did Lagos believe in? The relativist or the universalist?"

"He did not seem to think there was much of a difference. In the end, they are both somewhat mystical. Lagos believed that both schools of thought had essentially arrived at the same place by different lines of reasoning."

"But it seems to me there is a key difference," Hiro says. "The universalists think that we are determined by the prepatterned structure of our brains -- the pathways in the cortex. The relativists don't believe that we have any limits."

"Lagos modified the strict Chomskyan theory by supposing that learning a language is like blowing code into PROMs -- an analogy that I cannot interpret."

"The analogy is clear. PROMs are Programmable Read-Only Memory chips," Hiro says. "When they come from the factory, they have no content. Once and only once, you can place information into those chips and then freeze it -- the information, the software, becomes frozen into the chip -- it transmutes into hardware. After you have blown the code into the PROMs, you can read it out, but you can't write to them anymore. So Lagos was trying to say that the newborn human brain has no structure -- as the relativists would have it -- and that as the child learns a language, the developing brain structures itself accordingly, the language gets 'blown into' the hardware and becomes a permanent part of the brain's deep structure -- as the universalists would have it."

"Yes. This was his interpretation."

"Okay. So when he talked about Enki being a real person with magical powers, what he meant was that Enki somehow understood the connection between language and the brain, knew how to manipulate it. The same way that a hacker, knowing the secrets of a computer system, can write code to control it -- digital nam-shubs?"

"Lagos said that Enki had the ability to ascend into the universe of language and see it before his eyes. Much as humans go into the Metaverse. That gave him power to create nam-shubs. And nam-shubs had the power to alter the functioning of the brain and of the body."

"Why isn't anyone doing this kind of thing nowadays? Why aren't there any nam-shubs in English?"

"Not all languages are the same, as Steiner points out. Some languages are better at metaphor than others. Hebrew, Aramaic, Greek, and Chinese lend themselves to word play and have achieved a lasting grip on reality: Palestine had Qiryat Sefer, the 'City of the Letter,' and Syria had Byblos, the 'Town of the Book.' By contrast other civilizations seem 'speechless' or at least, as may have been the case in Egypt, not entirely cognizant of the creative and transformational powers of language.' Lagos believed that Sumerian was an extraordinarily powerful language -- at least it was in Sumer five thousand years ago."

"A language that lent itself to Enki's neurolinguistic hacking."

"Early linguists, as well as the Kabbalists, believed in a fictional language called the tongue of Eden, the language of Adam. It enabled all men to understand each other, to communicate without misunderstanding. It was the language of the Logos, the moment when God created the world by

speaking a word. In the tongue of Eden, naming a thing was the same as creating it. To quote Steiner again, 'Our speech interposes itself between apprehension and truth like a dusty pane or warped mirror. The tongue of Eden was like a flawless glass; a light of total understanding streamed through it. Thus Babel was a second Fall.' And Isaac the Blind, an early Kabbalist, said that, to quote Gershom Scholem's translation, 'The speech of men is connected with divine speech and all language whether heavenly or human derives from one source: the Divine Name.' The practical kabbalists, the sorcerers, bore the title Ba'al Shem, meaning 'master of the divine name.'"

"The machine language of the world," Hiro says.

"Is this another analogy?"

"Computers speak machine language," Hiro says. "It's written in ones and zeroes -- binary code. At the lowest level, all computers are programmed with strings of ones and zeroes. When you program in machine language, you are controlling the computer at its brainstem, the root of its existence. It's the tongue of Eden. But it's very difficult to work in machine language because you go crazy after a while, working at such a minute level. So a whole Babel of computer languages has been created for programmers: FORTRAN, BASIC, COBOL, LISP, Pascal, C, PROLOG, FORTH. You talk to the computer in one of these languages, and a piece of software called a compiler converts it into machine language. But you never can tell exactly what the compiler is doing. It doesn't always come out the way you want. Like a dusty pane or warped mirror. A really advanced hacker comes to understand the true inner workings of the machine -- he sees through the language he's working in and glimpses the secret functioning of the binary code -- becomes a Ba'al Shem of sorts."

"Lagos believed that the legends about the tongue of Eden were exaggerated versions of true events," the Librarian says. "These legends reflected nostalgia for a time when people spoke Sumerian, a tongue that was superior to anything that came afterward."

"Is Sumerian really that good?"

"Not as far as modern-day linguists can tell," the Librarian says. "As I mentioned, it is largely impossible for us to grasp. Lagos suspected that words worked differently in those days. If one's native tongue influences the physical structure of the developing brain, then it is fair to say that the Sumerians -- who spoke a language radically different from anything in existence today -- had fundamentally different brains from yours. Lagos believed that for this reason, Sumerian was a language ideally suited to the creation and propagation of viruses. That a virus, once released into Sumer, would spread rapidly and virulently, until it had infected everyone."